

Live SITL drift: root cause and fix

Why every live flight in the app slid slowly backwards while the same flight headless held still, how it was traced, and what changed. The design results (foil and jet sweeps, CFD calibration) ran headless and were never affected.

Cause. The app's simulator clock kept running from flight to flight. In SITL lockstep a freshly booted PX4 takes its clock from the first sensor message, so every flight after the first booted PX4 hundreds of seconds in. PX4 booted at a large time loses attitude accuracy: it missed about 1° of the nose-lift rotation and then hovered with a steady 0.3-0.4° nose-up lean, which pushed it backwards.

Fix. The sim clock restarts at zero before every SITL PX4 launch (`Simulator.restart_clock()`). Live Altitude-mode drift fell from 1.1-3.2 m to 0.3-0.7 m, Position mode from 0.4 m to 0.02-0.07 m, estimate error from 0.3-0.4° to 0.00°.

What was seen

Park -10 design (jets 57.5 / 85 / 85, hover 8.5°). In Stabilized and Altitude mode the live aircraft slid backwards at a steady ~0.2 m/s; Position mode held, because the H-FLOW sees the slide and PX4 leans against it.

FLIGHT, 60 S HOVER	HEADLESS	LIVE (BEFORE)
Stabilized drift	3-6 m	12-45 m
Estimate vs truth (pitch)	0.04-0.1°	0.3-0.4°
Rotate-up tracked by the estimate	99-101 % (26 of 26 runs)	87-95 % (9 of 10 runs)

How it was traced

- 1

Not a gain problem.

PX4 tracked its own attitude target within 0.01°. The error sat one step earlier: its estimate was 0.3-0.4° more nose-down than the truth. No integrator can see that, so raising I would have changed nothing. located
- 2

Where it appears.

During the on-ground rotate-up the estimate lost about 1° in one go, ~5 s in, as the airframe passed level, then kept the offset for the whole flight. Live: 9 of 10 runs. Headless: none of 26. located
- 3

Ruled out, one at a time:

PX4's stock simulation settings (3 estimators, gyro rate, FFT, autotune, GPS delay), boot logging, pre-flight settling, the fan-vibration model, real-time pacing, 1000 Hz physics, home position, a clean seeded PX4 boot. The sim loop was healthy: zero lockstep timeouts, real-time factor 1.00. not it

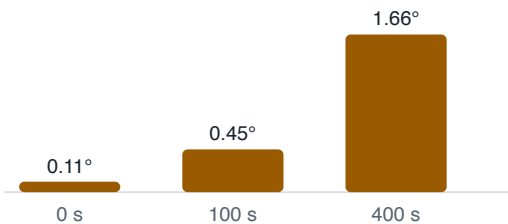
4 The pattern.

The only clean live flight was the first after an app restart, with the clock at 16 s. The bad ones started at 88-1,100 s. Headless runs always start at 0. **hypothesis**

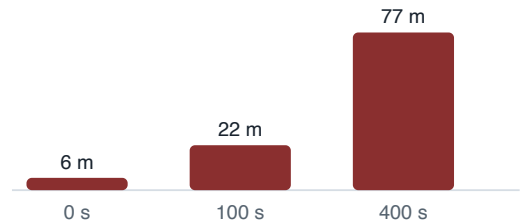
5 Reproduced headless

by starting the clock late, nothing else changed: **confirmed**

Estimate error in hover (deg) vs clock at PX4 boot



Stabilized drift in 60 s (m) vs clock at PX4 boot



Verified live after the fix

Two Altitude → Position → Altitude flights back to back (the second is the real test: before the fix every flight after the first inherited the clock).

	BEFORE	AFTER, FLIGHT 1	AFTER, FLIGHT 2
Rotate-up tracked	93-94 %	100.3 %	100.1 %
Estimate vs truth	0.26-0.42°	0.01°	0.00°
Altitude-mode drift (hold / 10 s / landing)	1.1-1.3 / 1.6-1.9 / 2.7-3.2 m	0.67 / 0.28 / 0.57 m	0.63 / 0.41 / 0.59 m
Position-mode drift, 60 s	0.36-0.40 m	0.07 m	0.02 m
Touchdown	0.37-0.40 m/s	0.24 m/s	0.24 m/s

Other live-app fixes on the way

- **Fly live no longer stalls.** A relaunched PX4 sometimes came up with its estimator starved (only Manual / Acro armable). The app now checks the estimator after each fresh start and relaunches up to 3 times, with a clear message if it never starts.
- **Clean, seeded PX4 boot** for every live flight, like a headless run (airframe and scenario parameters written in before boot).
- **Settle gate:** a flight waits until PX4's attitude estimate agrees with the truth on the legs before rotating.
- **Batch and Tuning tabs** reload the scenario list each time they open, so newly saved scenarios appear without a page refresh.

HITL: separate, still open

The board is not affected by this bug: it timestamps sensor data on arrival with its own clock. HITL on the park -10 design already matched live SITL (Position hold 0.42 m vs 0.39 m), but it carries its own $\sim 0.45^\circ$ estimate offset, and the board's log shows “Found 0 compass” with the estimate 2.4° off at rest. That points at the board not receiving or accepting the simulated magnetometer, and is the next thing to check.

Files

`airframe_designer/sim/simulator.py` (`restart_clock`), `airframe_designer/px4/connection.py` (clock restart and seeded boot on launch, `estimator_up`), `airframe_designer/server/app.py` (estimator retries, seeding), `airframe_designer/sim/scenario.py` (settle gate), `ui/app.js` , tests in `tests/test_sim.py` (262 pass). Evidence: `results/v3_30/rootcause/` (headless clock and loop variants), live runs `results/tuning/live9*.json` . Commits `d3c3748`, `332fd34`. Brain: `brain/lessons/live-app-needs-fresh-px4-per-flight.md` .

ATLAS V3_30 simulator, PX4 1.17 SITL lockstep, 250 Hz sensors. Generated 2026-10-01.